

WILL Relational Geometry

Open Research in Foundations of Physics

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Living Research Protocol

Abstract

By applying extreme methodological constraints:

- **Epistemic Hygiene:** Refusal to import unjustified assumptions.
- **Relational Origin:** Physical quantities must be defined by their relations. No background allowed.
- **Ontological Minimalism:** Minimum foundational primitives.
- **Mathematical Transparency:** No ontologically vague mathematics allowed.

we build theory without new postulates, free parameters or arbitrary choices. Laws are not introduced to fit observations. They emerge as necessary consequences of geometric self-consistency. We recover all main results of Special and General Relativity, derive atomic structure, and extend to cosmology and galactic dynamics. All results are tested against multiple datasets with zero adjustable parameters:

- S2 star orbit around Sgr A* (combined astrometric + spectroscopic data 24 years)
- SPARC galaxy rotation curves (175 galaxies)
- Planck 2018 CMB acoustic spectrum
- Pantheon+ Type Ia supernovae
- Gaia DR3 wide binaries

IMPORTANT: LIVING DOCUMENT PROTOCOL

This is a static Gateway Document.

WILL Relational Geometry is continuously updated research. To prevent outdated files and ensure version control, full manuscripts are hosted exclusively as live documents. This PDF exists solely for academic platform indexing.

Access the current WILL Trilogy here:

- **Part I: Relational Geometry** willrg.com/documents/WILL_RG_I.pdf
- **Relational Orbital Mechanics (R.O.M.)** [willrg.com/documents/WILL RG R.O.M..pdf](https://willrg.com/documents/WILL_RG_R.O.M..pdf)
- **Part II: Relational Cosmology** willrg.com/documents/WILL_RG_II.pdf
- **Part III: Relational Quantum Mechanics** willrg.com/documents/WILL_RG_III.pdf

Project Resources & Live Links

- ◇ **Main Hub:** willrg.com
- ◇ **Documents and Results:** willrg.com/results
- ◇ **Falsifiable Predictions:** willrg.com/predictions
- ◇ **WILL-AI:** willrg.com/will-ai
- ◇ **GitHub Repo:** github.com/AntonRize/WILL
- ◇ **YouTube:** youtube.com/@anton_rize

*Archived on Zenodo: <https://zenodo.org/record/17115270>

A Direct Note to the Reader: Why engage with this research?

Engaging with unknown independent research is a substantial commitment, and you should not do it blindly.

I recommend the following: Be honest with yourself. Define the exact criteria this research must satisfy to justify the investment of your time. Then, ask [WILL-AI](#) if this work meets your standards. The AI is trained on this papers and will work as advance search engine. It will give you an honest, unbiased answer, within its limitations, and will provide direct links to related sections in the trilogy, ready to run colab notebooks and desmos projects. If you are still in doubt, browse willrg.com:

- **Falsifiable Predictions** - from atomic structure through local neighbourhood to cosmic horizon;
- **Galactic Dynamics** - test rotation curves predictions against **SPARC dataset**;
- **Holographic Decoder** - **Breaking $M \sin(i)$ Degeneracy** with 1D RV data string;
- **Computational Efficiency** - side by side comparison with General Relativity;
- **Logos Map** - trace the derivation logical chain;
- **WILL-AI** - just have a chat its fun.

Regardless of your decision, I wish you to find what you are seeking.

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